Calibration

1. Connect the tube from the reservoir C (NaI) to the source.
2. File -> open Global/Local HDX file. Check that all the parameters are correct in MS Tune: Acq Mode -> Positive/Negative Ion, Setup -> Quad Profile -> Autoprofile -> Update, Instrument parameters and Fluidics. Flow state -> infusion, Reservoir -> C, Sprayer Position -> Sample.
3. Start infusing and acquiring, file name Date_Name_Mode (usually Res) _mass_range (e.g. _100_2000).
4. When the signal becomes stable ~e5, right click and drag on the chromatogram to get a combined spectrum. Save it, stop infusing and acquiring.
5. Spectrum -> Process -> Centre -> Okay, save the centred one as well. This one is used for calibration.
6. MS Console -> Synapt Gi 2 -> IntelliSTART -> Create Calibration -> Start. In case of some problems try choosing Normal Mode -> Start -> Uncal -> switch to Configuration Mode.
7. Calibration Profile Editor -> File -> New. There choose Manual Mode, Resolution, Compound Sodium Iodide (positive). For the file go in History -> choose AccMass, this is the centred spectrum.
8. Finally, check that 13/13 peaks were recognised, RMS Residual Mass is < 5 ppm. If so, accept the calibration (top left corner, green tick).
9. Switch the wire to the one from the LC.

If the calibration is not good enough (for example, for Native MS m/z range100-5000 calibration only 21 out of 33 peaks are detected), you can try to change the Quad profile to Manual instead of auto and set the interval 100-2500-5000. It might help, but if not as long as the RMSD is less than 5 ppm you can just accept it.

LC MS + HDX

First step is to install the columns, because it will take time to cool down the chamber back to 0 degrees.

All measurements should start with the blank experiment (press Inject button without actually injecting anything) to check that the columns are installed correctly and the system is ready. The masses can be checked based on 3-4 peaks using ESIprot Online website.
Sprayer position should be changed to Lockspray.

Before the experiment, wash the syringe 3 times with Strong Wash, 3 times with Weak Wash. After each injection (after pressing the button):

Wash the syringe 3 times with Strong Wash
Inject Strong Wash 3 times
Wash the syringe 3 times with Weak Wash
Inject Weak Wash 3 times

Problem solving

In case the problem is in the software/instrument connection, try Inlet Method -> LC -> Reset Communication.
If it doesn't help restart the MassLynx.
In case if restarting MassLynx doesn't help to solve the problem, try restarting the computer.

M+31 -> check if the bottle for the mobile phase used to contain Methanol (32 Da), rinse it properly and prepare the solution again.

Preparation

Before starting the experiment check that all the necessary solutions are full: lockspray, calibration, liquid phase for LC (both A and B), buffers (H2O and D2O), and quenching solution are prepared. If you change any LC solution or Lockspray, purge the system first.

If quenching solution or buffers are new, you have to check the pH value using pH paper (pH ~ 2.5) first by mixing them in the same ratios as in the sample.

5 uL H2O buffer + 5 uL quenching solution => measure pH
1 uL H2O buffer + 9 uL D2O buffer + 10 uL quenching solution => measure pH

If the LC liquid phase is new as well, purge the system to wash the external parts with the new solvent:

Mass Lynx -> MS Console -> UPLC
Binary Solvent Manager -> Control -> Prime A/B Solvents -> A and B 5 min
Auxiliary Solvent Manager -> Control -> Prime A/B Solvents -> A 5 min

Before starting the experiment you have to turn on the flow and wait for the pressure to stabilise. It should be ~6000 psi, $\Delta < 20$.
2 methods: press uL/min and change manually ***OR*** LC -> Load Method (previously open the necessary file through Open -> Lena -> ACQUDB -> …). Typically BSM 60 uL/min, ASM 100 uL/min.

Next create the queue of experiments and browse MS and Inlet files for them. Select the experiments you want to perform and click Start. When the phrase "Waiting for injection…" appears, you can start the injection manually by pressing the white button.

General advice regarding concentrations

For Global the amount of protein in the injection should be < 10 pmole
For Local the amount of protein in the injection should be < 50 pmole
Signal should be ~e6, carryover should not exceed 10%.

Columns

For Local: C18 + pepsin column, For Global: C4

HDX

1. Prepare the quenching solution (30 uL of 1.5% FA) in the eppendorf tubes and put them in the ice (open).
2. Add 3 uL of the protein in the tube and prepare 27 uL of D2O buffer in the pipette.
3. Set the timer for xx:10 (number of minutes needed + 10 seconds). When it gets to xx:00 mix D2O in with protein.
4. Change the volume of the pipette to 30 uL and wait for 00:10. Then take the solution with the pipette and mix it with the quenching solution (still in the ice) at 00:00.

Global HDX

Normal pressure ~5000 psi, ASM ~700
To wash the columns: Backflush OR Solvent B 70% for 1-2 hours, then Solvent B 5% for 30 minutes and do the blank again.

Local HDX

Normal BSM pressure ~12500 psi, ASM ~700 psi

For local HDX it is crucial to check the lockspray solution signal!!!
You can check the real-time signal in the Mass Tune page by switching the sprayer position to Lockspray if necessary. Throughout the measurement you can check the lockspray signal in Chromatogram -> Display -> TIC -> 2nd or 3rd option. Lockspray infusion rate 2 uL/min

Strong and weak wash can be used right after the injection. Wash the pepsin column with the pepsin wash after the trapping is over, then press the button Wash for ~10 seconds 3 times. After the control (which should be checked for coverage by PLGS + DynamX) the carryover should be checked. If PLGS is unable to find any peptides in the carryover measurement, you can skip this step and just perform injections with the sample without blanks.

Native MS
Buffer exchange and concentrations

Buffers such as PB/PBS/TRIS are not suitable for nano-ESI. The most compatible one is ammonium acetate solution or bicarbonate (trypsin digestion), 10-50 mM. This source can also tolerate AcN and MeOH.

Cut-off value of the buffer exchange device is 10k. Put 20 uL of stock protein solution into the buffer exchange device + fill up completely with the buffer for exchange (e.g. ammonium acetate solution) ~480 uL. Centrifuge at the maximum speed (16800 rpm) for 5 min 2 times and for 15 min last time. In the best case if you have enough time do 15 min each. For BL samples 3 rounds of buffer exchange don’t seem enough, better do 6 times.

The concentrations before and after can be checked with nanodrop, but in general the dilution is ~0.5. The final concentration of the protein for native MS should be ≤ 10 uM, but it depends on the signal.

Experiment

Source Temperature 40 degrees, it takes hours to cool down, so make sure you change the source in advance (ask the engineer). Calibration interval 100-5000.

First open the needle using the tweezers and the microscope. Try not to open it too much, otherwise the sample will run out too fast and you won't be able to collect enough information. Make sure all 3 parts are present in the needle set up (including a small rubber part). To move the source and insert the needle, unscrew the source first, then press the button on the side to let it move.

To make sure you detect all the signals from the sample, start acquiring first with 0 voltage and 0 flow gas. Then turn on the voltage to 1.5 kV and nano flow gas to 0.15 Bar. Once there's flow (you can see the signal in MS window) stop the gas flow to slow down the sample introduction. If the signal is unstable, you can try to increase the nano flow gas up to 0.5 and quickly switch it back to 0.2 when the signal improves.
For needles coated with 2 nm gold the following parameters showed a good result: Source Temperature 80, Capillary 2 kV, Sampling Cone/Source Offset 30, Cone Gas 60 L/h, 0.5 bar nanoflow gas; additionally to evaporate more solvent and reduce signal to noise ratio Trap Collision Energy 15, Transfer Collision Energy 5, Gas Controls 1. Trap 2.0, 2. Helium Cell 180, IMS 90

Before changing the needle, turn off the voltage and the instrument for safety.

Needles

Usually borosilicate glass rods with o.d. 1 mm are used. Firstly, align ~20 of them and cut them in half using a ceramic knife. Turn on the machine and choose program 96 -> Enter. Insert the glass rod in the left holder so that it goes through the hole in the middle, then screw the holder. Shift it to the middle, screw another holder and close the lid. Press Pull and see 2 needles formed. After you fill the whole container with needles, put them in a gold coating machine. Before running the program check that the container can rotate freely inside. Then use the pen to choose the QT Gold 5 nano file or Kin's method.

Ion Mobility

Change the Acquisition Mode -> Mobility TOF. To record the drift time data conditions of the experiment remain the same as for the Native MS, but when you press Acquire -> don't forget to tick "Add drift time function". To display drift time use Display -> Mass -> Function -> select 2nd channel, pick the m/z range of interest. Higher charge states have lower drift times.

Adduct Ions

Cationized Adducts		Anionized Adducts	
$[M+NH_4]^+$	$M_r + 18$	$[M+Na]^-$	$M_r + 21$
$[M+Na]^+$	$M_r + 23$	$[M+Cl]^-$	$M_r + 35$
$[M+CH_3OH+H]^+$	$M_r + 33$	$[M+HCOO]^-$	$M_r + 45$
$[M+K]^+$	$M_r + 39$	$[M+OAc]^-$	$M_r + 59$
$[M+CH_3CN+H]^+$	$M_r + 42$	$[M+K]^-$	$M_r + 37$
$[M+DMSO]^+$	$M_r + 79$	$[M+TFA]^-$	$M_r + 113$
Preformed		**Preformed**	
M^+; $[M+H]^+$	M_r ; $M_r + 1$	M^-; $[M-H]^-$	M_r ; $M_r - 1$

Solutions for experiment

Ni trocefin test

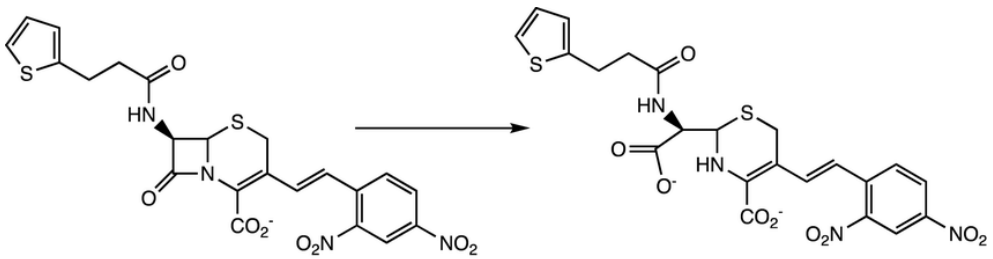
1. Prepare 5 ug/uL stock solution of nitrocefin (516.5 g/mole) in DMSO (colour: deep red)

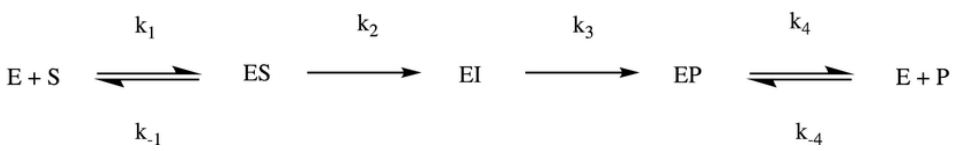
   1.5 mg of nitrocefin powder (orange) + 300 uL DMSO

2. 1.7 uL of stock solution + 48.3 uL of 50mM PB (pH 7.4) => pale yellow

   (Even if the solution looks too diluted and colourless, the test might work. If it doesn't, first try to increase the nitrocefin concentration, second - replace the stock solution, it should be fresh.)

3. Cut a small piece of parafilm and add 5 uL of diluted nitrocefin solution onto it, forming a droplet.
4. Add 5 uL of the enzyme to that droplet and observe the reaction - solution should turn pink/red in around 5 minutes. Concentration of the solution ~2 uM.





Although nitrocefin is considered a cephalosporin, it does not appear to have antimicrobial properties. (O'Callaghan 1972) The absorption wavelength shifts from intact (yellow) nitrocefin (~380 nm) to degraded (red) nitrocefin (~500 nm).

In BL assays chapter stock is prepared in the following way:

5. Prepare stock solution of nitrocefin (516.5 g/mole) by dissolving 10.3 mg in 0.5 ml DMSO

6. Add desired buffer to the stock to reach 10 ml (DMSO 5%), this solution (2 mM) is stable for several weeks at 4 C

Phosphate buffer

K2HPO4 174.18 g/mole
KH2PO4 136.09 g/mole

D2O density => 10g of D2O = 9.055 mL
pD = pH + 0.4

1. 100 mL of 50 mM PB, pH 7.4

50 mL of 50 mM K2HPO4 50 mM * 50 mL = 0.0025 mole => 0.0025 * 174.18 = 0.4355 g
50 mL of 50 mM KH2PO4 50 mM * 50 mL = 0.0025 mole => 0.0025 * 136.09 = 0.3402 g

Mix and titrate with Na OH

2. 9.055 mL of 50 mM D2O, pD 7.4

4.5275 mL of 50 mM K2HPO4 50 mM * 4.5275 mL => * 174.18 = 0.0394 g
4.5275 mL of 50 mM KH2PO4 50 mM * 4.5275 mL => * 136.09 = 0.0308 g

To adjust the pH and pD, the pH-meter needs to be calibrated first: turn on, press "cal", measure the reference solution, press "read"; for the measurement itself press "read".

For H2O buffer to adjust pH to 7.4 around 130 uM of 6M NaOH have to be added.
For D2O buffer to adjust pD to 7.4 (pH 7.0!!!) around 1uL of concentrated FA has to be added.

AA solution

Ammonium Acetate 77.08 g/mole

15 mL * 20 mM = 300 umole => 300 umole * 77.08 = 0.02312 g

Solutions for MS

mQ water R = 18.2 MΩ, not lower.

LC A solution

0.1% FA (400mL H2O + 400 uL 100% FA)

LC B solution

AcN with 0.1% FA (400 mL AcN + 400 uL 100% FA)

Weak wash

0.1% FA

Strong wash

H2O:AcN:MeOH:IPA

1:1:1:1

Pepsin wash

8M GuHCl 3.75 mL

AcN 0.8 mL

FA 0.16 mL

H2O ~15 mL

LE (Leucine enkephalin, 556.2771 Da)

100 uL LE std

2.5 mL AcN

2.5 mL H2O

5 uL FA

=> final LE concentration ~ 8 ng/uL

Stock LE: 3 mg LE + 7.5 mL H2O ~400 ng/uL

NaI

1 mg/mL in MeOH

Data Analysis

Parameters

To check the parameters of the source/inlet method used in the experiment, you can go to MassLynx -> Chromatogram, Open-> pick a file, but don't open it! Instead, press Experiment and save the file with all parameters.

Mass plot

1. If needed the data can be smoothed (channels 5, number 3) and the baseline subtracted.
2. Centre the spectrum: min peak width 3, Top.
3. Process -> Component -> Find Manual. Then right click on the highest peak and an adjacent one. -> Calculate.
4. Process -> Component -> Edit. Now we can include or reject peaks manually, e.g. only keep the highest and ±5 peaks. OR keep them all. If some peaks are not detected decrease the threshold to ~ 2% BPI
5. Process -> Transform

Before calculating the mass, internal calibration can be performed using Lockspray.
Display -> TIC -> Channel 2 -> Combine -> Centre
Tool -> Lock Mass -> Enter the value -> Accept
Tools -> Modify Calibration -> Copy the value -> Paste it in Modify Calibration of another spectrum

To plot the uptake in percentages take into account 2 points: first AA is not an exchange we're tracking, so for all the proteins/peptides the (number of exchangeable amide protons) equals (overall number of residues - 1). If there are any prolines in the sequence they have to be subtracted as well.

Global HDX

Global HDX analysis can be performed using DynamX, but practice shows that calculating the masses for all exposures and making the uptake plot manually gives better results (check yourself).

Creating a library in PLGS

To create a library of peptides in PLGS, first of all you have to add a .fasta file with the sequence of your protein to C -> PLGS3.0.2 -> demodbs. Go to PDB -> Display files -> FASTA sequence -> Save as all files -> manually add .fasta.

1. Create a new project in PLGS.
2. Libraries -> Protein Sequence Databank -> New
3. Name the file, choose Standard-spaced format and upload the file in Location. Test the format of the file to make sure it is correct. -> Ok
4. Libraries -> Processing parameters -> New -> Electrospray MSe, Lock Mass Charge1 556.2771 Da. Actually, the processing file is the same for all proteins (no sequence in it).
5. Libraries -> Workflow parameters -> New. Choose Electrospray MSe, Digest reagent - non-specific (pepsin), unselect (Ctrl+Click) all modifiers.
6. Vials (Right Click) -> Add Vial
7. Right Click -> Add raw data
8. ~name -> Right Click -> Process
9. Double Click the name that has appeared

Created Excels are stored in C:\PLGS3.0.2\Export files\Ion accounting output*name*. You can check the location via Options -> Automation Setup.

If needed special modifications can be created in PLGS: Libraries -> Modifier Reagents -> Create all required modifications, then add them as variable modifiers in Workflow Parameters.

DynamX

For Global HDX in Data -> MS Files -> Experiment Type - Global Analysis.

For Local HDX PLGS files have to be browsed first. Use the control experiment to do the search and import the results into DynamX to check the coverage. You can upload multiple PLGS searches for different control experiments simultaneously to increase the coverage (set the File Threshold 0) or to improve the assignment of the peptides (set Threshold 2 for 3 controls for example)

Data -> Import PLGS files. Filtering depends on your data, e.g.
Min Intensity: 10 000
Min Products per Amino Acid: 0.3
Min Score: 7

After that a manual inspection is required. First for BLs go to Data -> Protein Editor -> Start numbering from 25. Views -> Stack Spectral Plot. Now we can check different charge states of all the peptides and check if they were assigned correctly. The calculation for the 1st peak is (m + N - 1)/N, where m is mass and N is the charge.

Shortcuts: Shift + Drag - assign peaks, Ctrl + Drag - Unassign peaks.

To get a peptide list go to Butterfly plot, Right Click -> Copy Data,paste it on a separate sheet in Excel and cut out the necessary columns.

Heat map -> Right Click ->Pymol
This command needs the protein to be named in order to work, so if no options appear got to Protein Editor and rename the protein.
Open Pymol structure and apply script
File -> Run…

After that just execute the command to colour the structure according to the heat map

spectrum b, blue_white_red, minimum=0, maximum=0.6

For State Comparison it's better to use other command

spectrum properties["CA - Default State 1 min"], blue_white_red, /1yt4, -0.2, 0.2
(/1yt4 - protein name, -0.2,0.2 - scale of spectrum)

spectrum properties["Default State 10 min"], blue_white_red, /1btl_tem1, 0, 0.5
spectrum properties["+CA 10 min"], blue_white_red, /1btl_tem1, 0, 0.5
spectrum properties["+TZB 10 min"], blue_white_red, /1btl_tem1, 0, 0.5